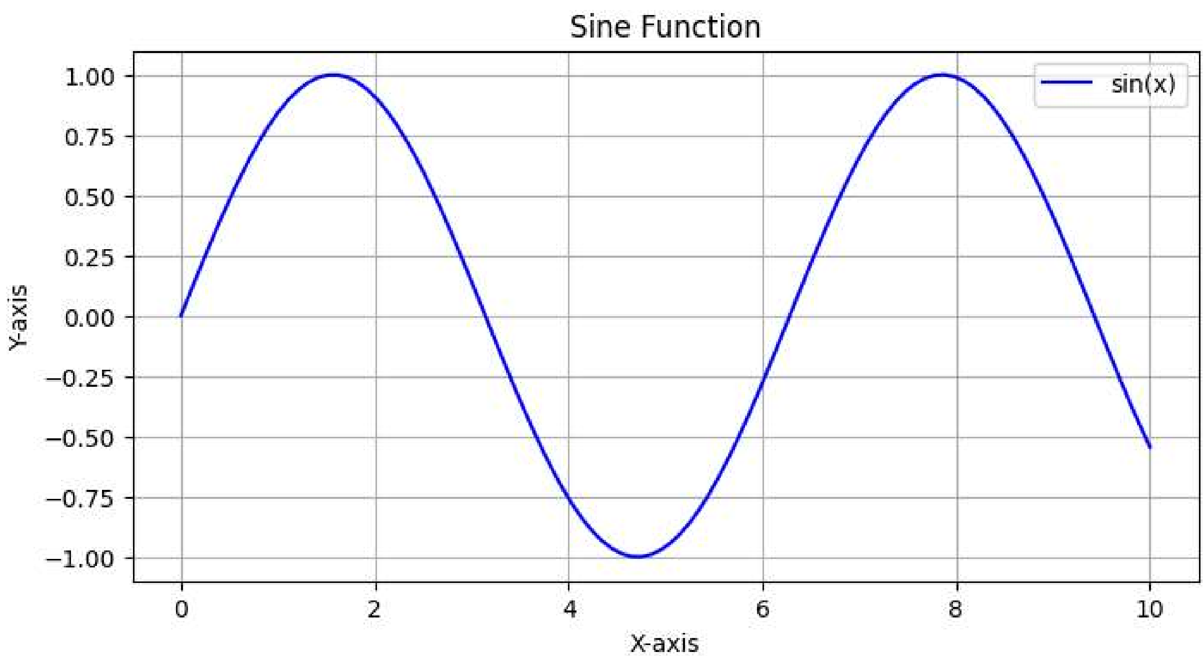
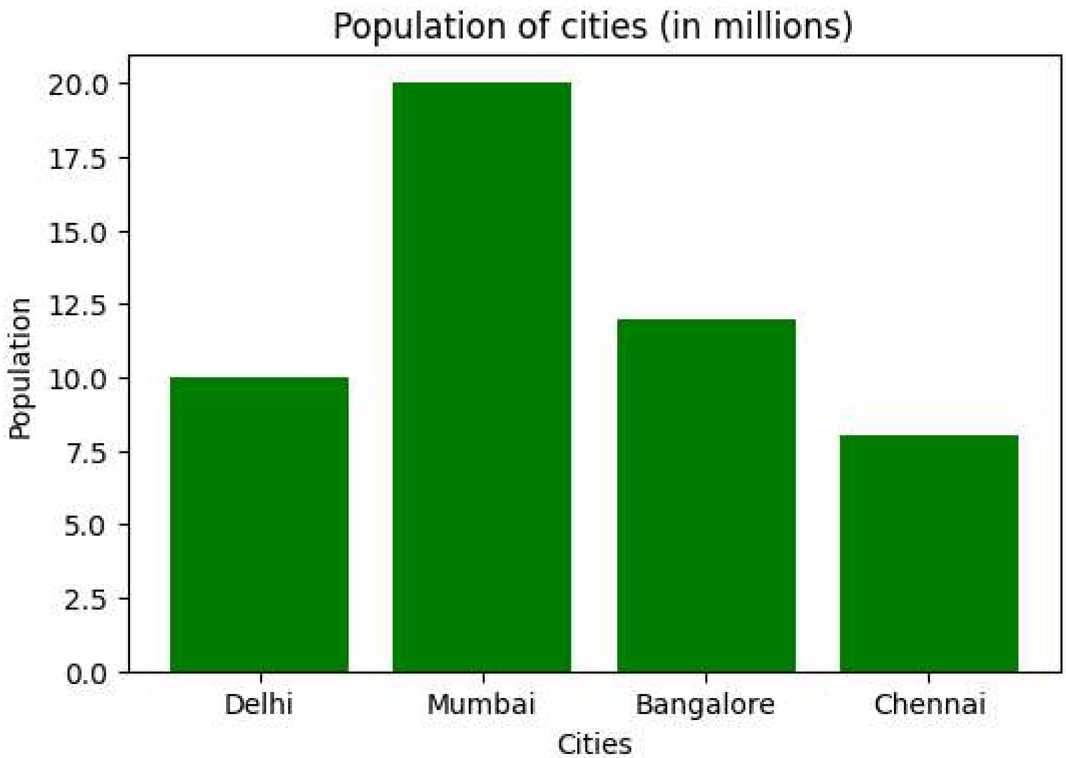
In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

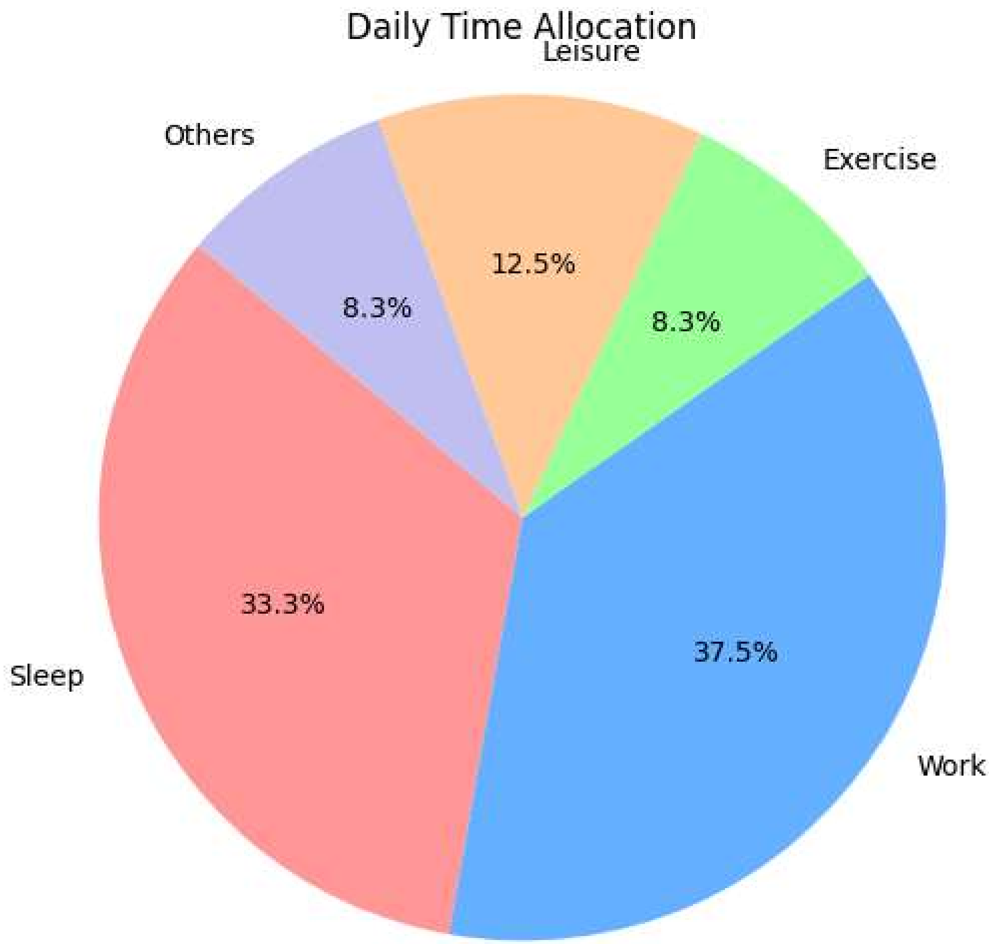
In [3]:
```
x = np.linspace(0,10,100)
y = np.sin(x)
plt.figure(figsize=(8,4))
plt.plot(x,y,label = "sin(x)",color='blue')
plt.title("Sine Function")
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.grid(True)
plt.legend()
plt.show()
```



In [4]:
```
cities =['Delhi','Mumbai','Bangalore','Chennai']
population = [10,20,12,8]
plt.figure(figsize=(6,4))
plt.bar(cities,population,color='green')
plt.title("Population of cities (in millions)")
plt.xlabel("Cities")
plt.ylabel("Population")
plt.show()
```

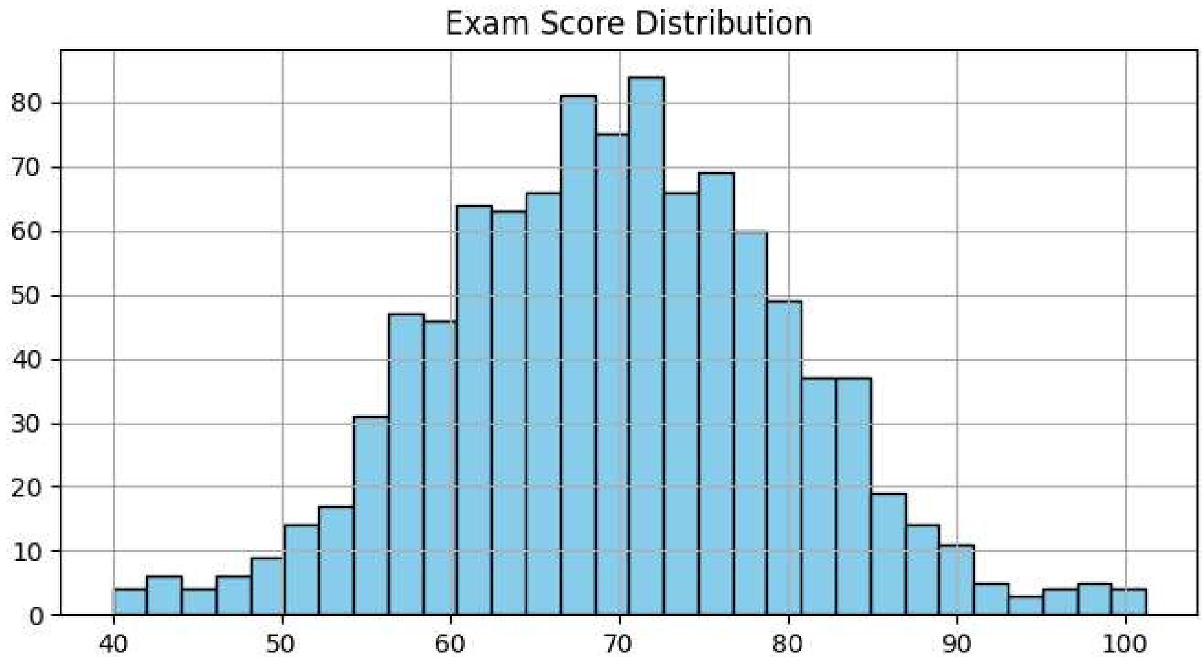


In [6]:
```
labels = ['Sleep','Work','Exercise','Leisure','Others']
sizes =[8,9,2,3,2]
colors = ['#ff9999','#66b3ff','#99ff99','#ffcc99','#c2c2f0']
plt.figure(figsize=(6,6))
plt.pie(sizes,labels=labels,colors=colors,autopct='%1.1f%%',startangle=140)
plt.title("Daily Time Allocation")
plt.axis('equal')
plt.show()
```



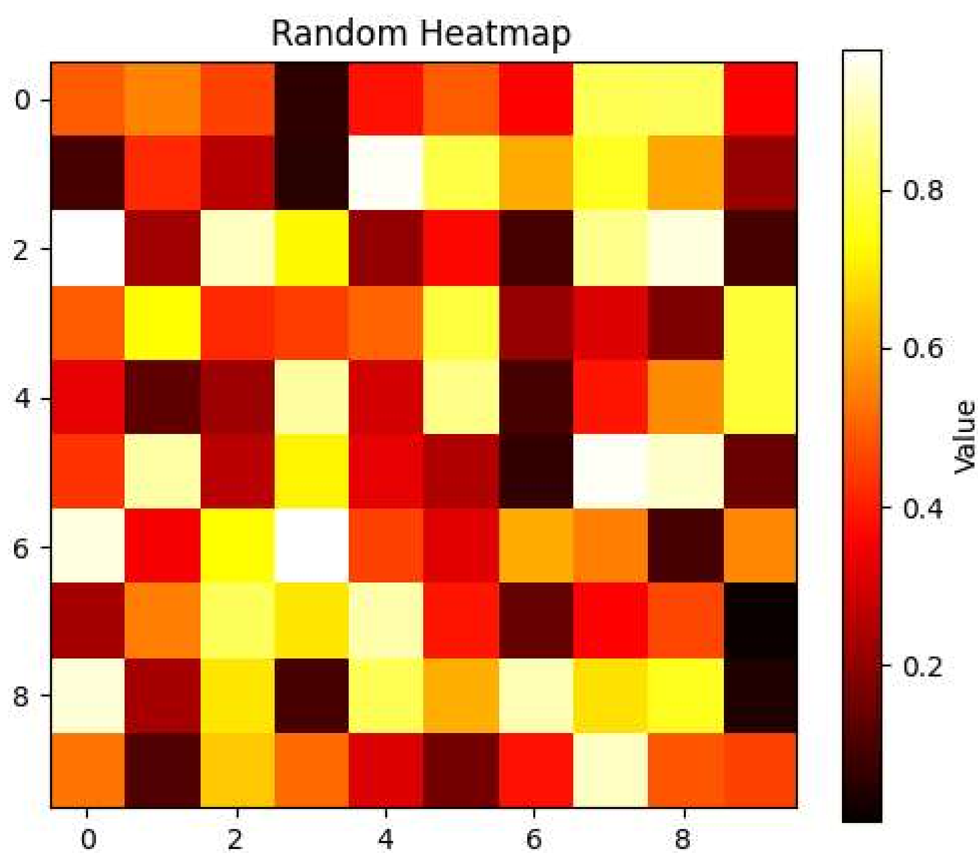
In [8]:

```
scores=np.random.normal(loc = 70,scale=10,size=1000)
plt.figure(figsize =(8,4))
plt.hist(scores,bins=30,color ='skyblue',edgecolor='black')
plt.title("Exam Score Distribution")
plt.xlabel = ('Score')
plt.ylabel = ('Number of students')
plt.grid(True)
plt.show()
```



In [12]:

```
data = np.random.rand(10,10)
plt.figure(figsize=(6,5))
plt.imshow(data , cmap='hot', interpolation ='nearest')
plt.colorbar(label='Value')
plt.title("Random Heatmap")
plt.show()
```



In [ ]: